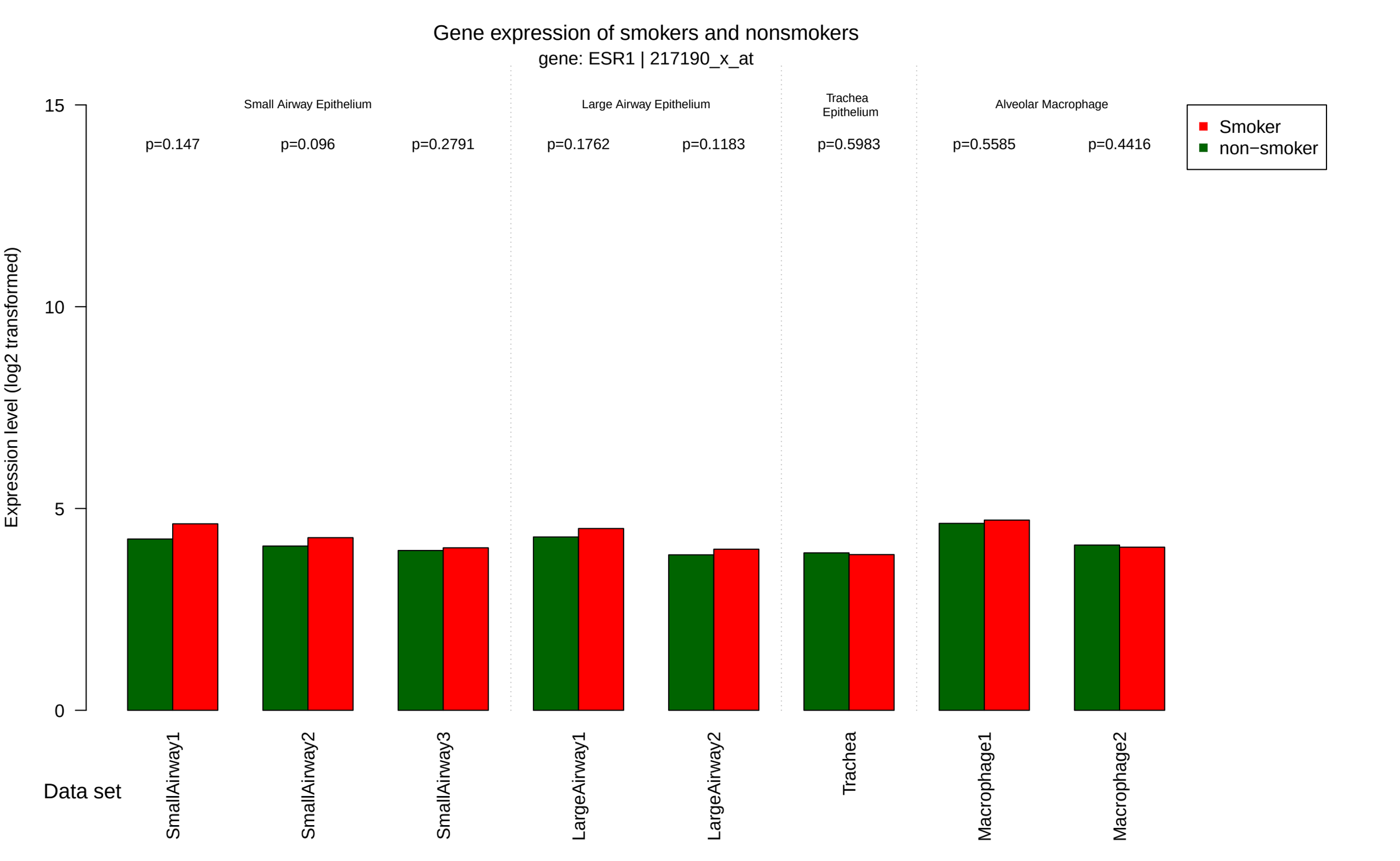
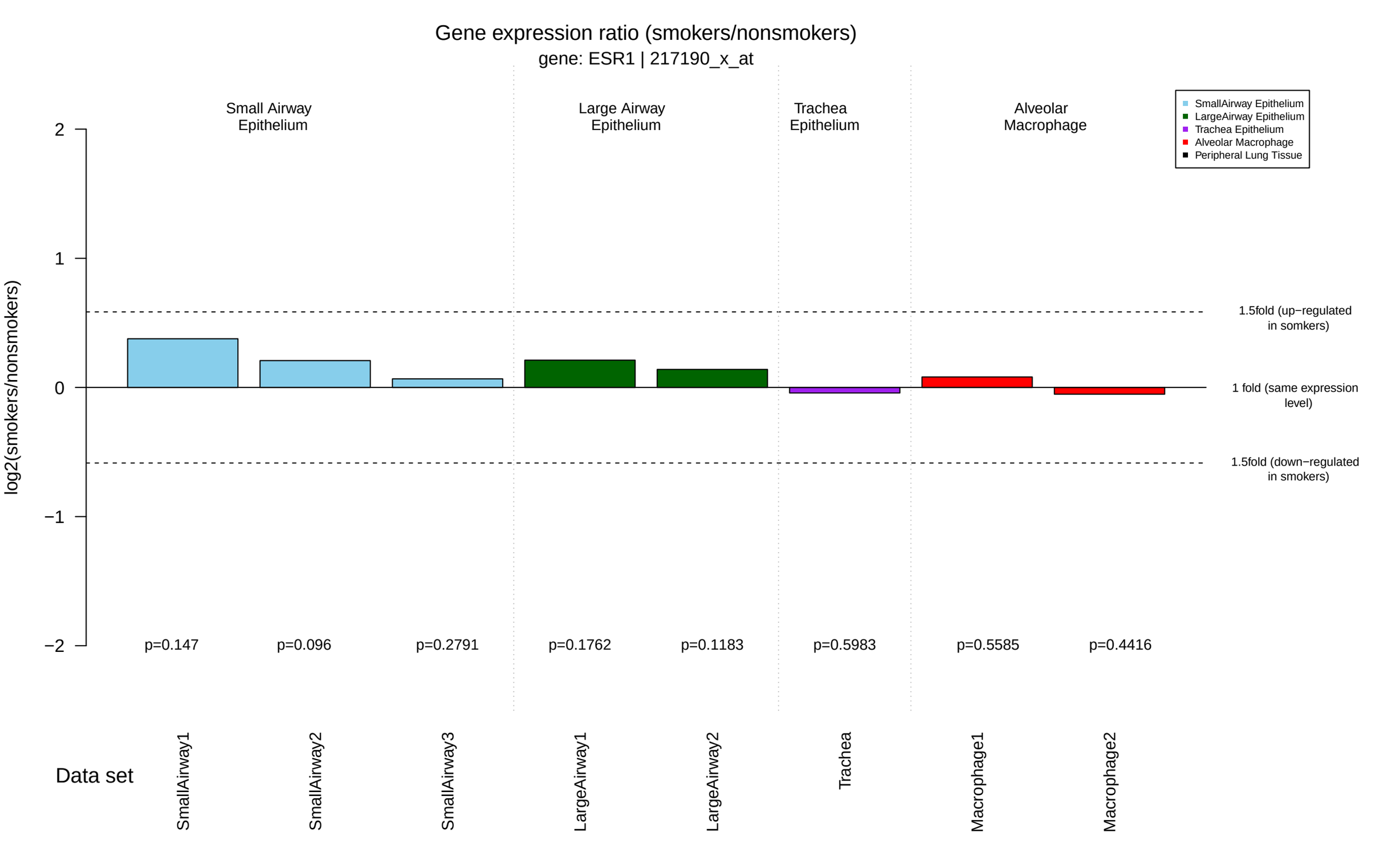
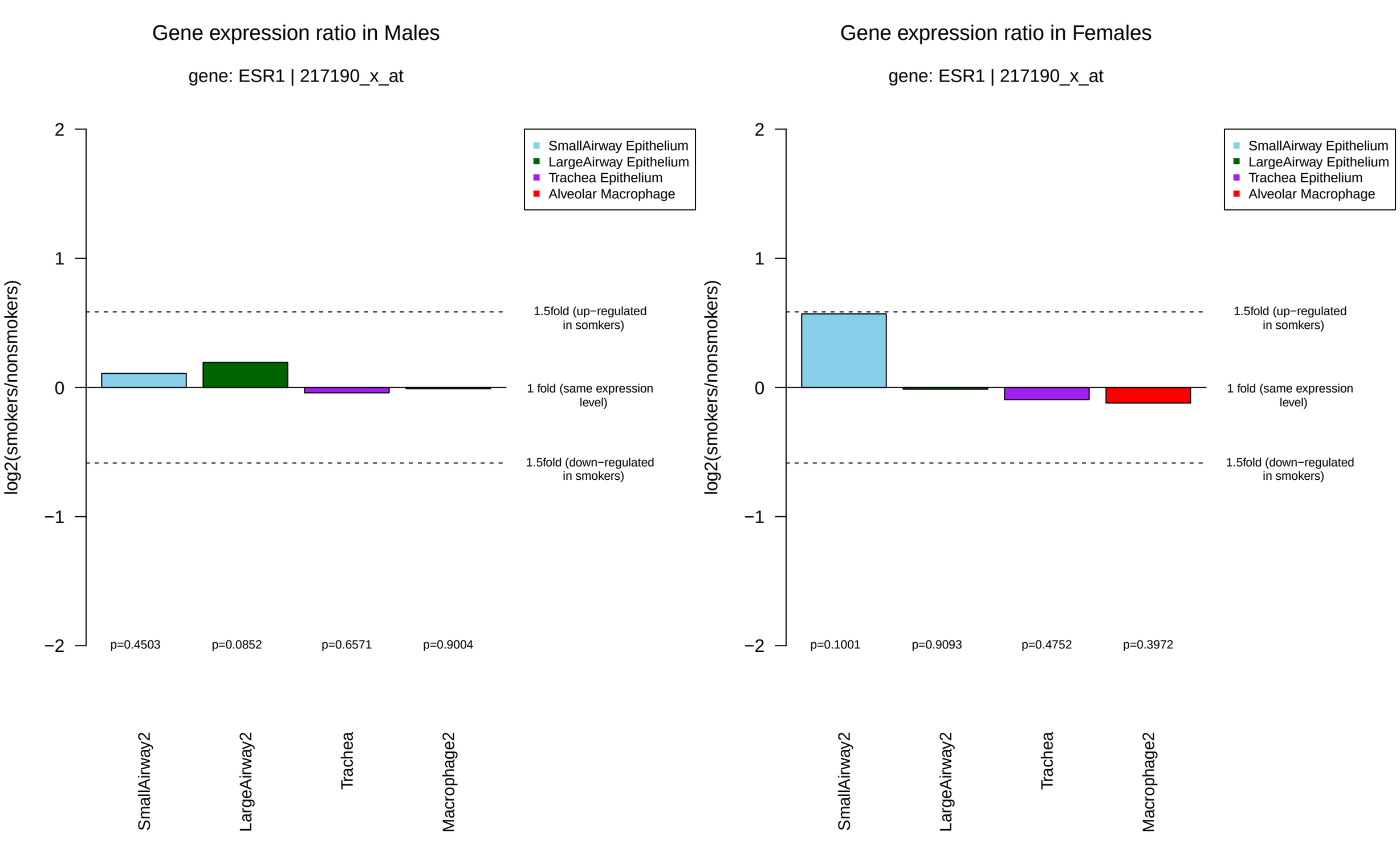
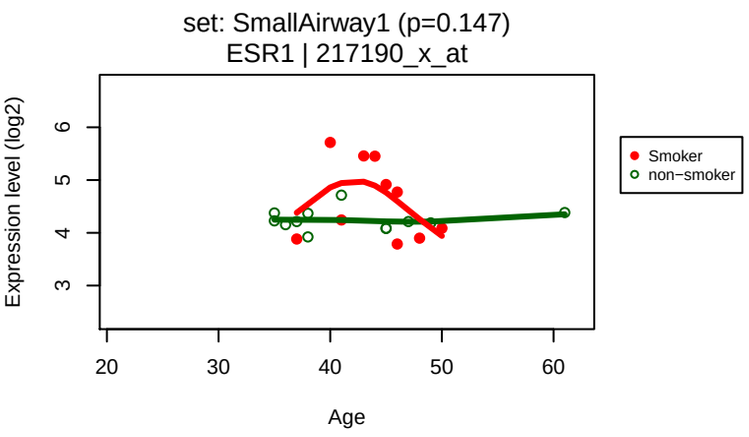
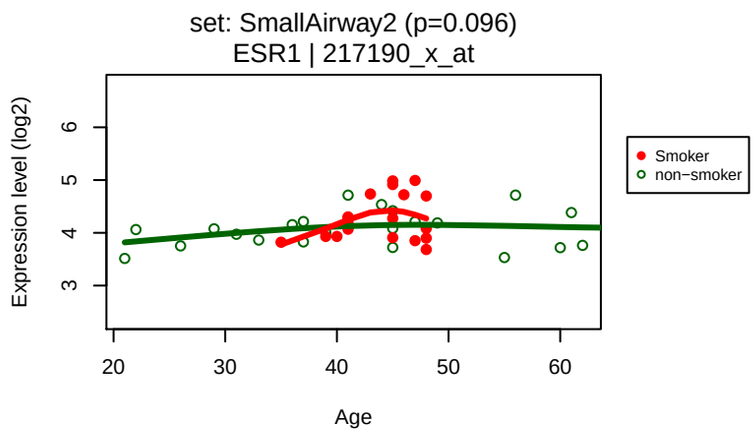
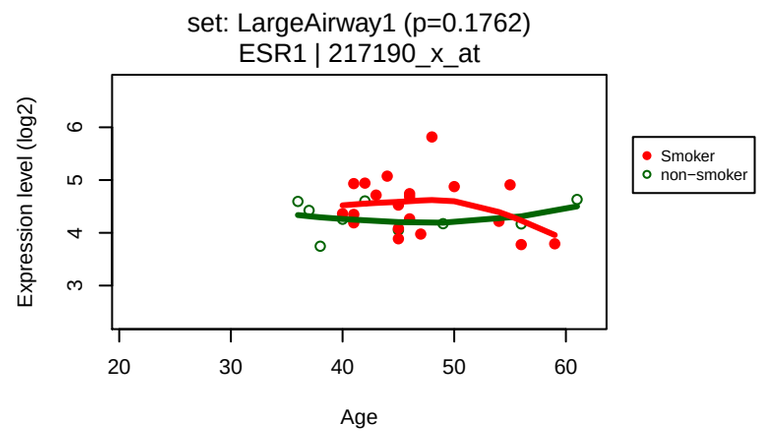

Gene expression of smokers and nonsmokers
gene: ESR1 | 217190_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.147
p=0.096
p=0.2791
p=0.1762
p=0.1183
p=0.5983
p=0.5585
p=0.4416
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

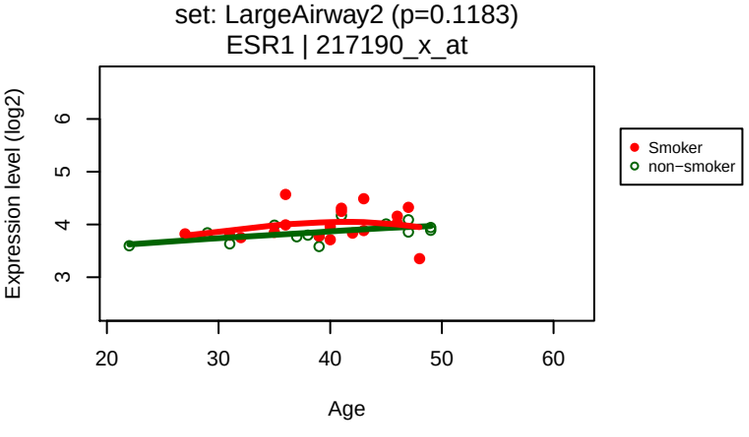

Gene expression ratio (smokers/nonsmokers)
gene: ESR1 | 217190_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.147
p=0.096
p=0.2791
p=0.1762
p=0.1183
p=0.5983
p=0.5585
p=0.4416
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

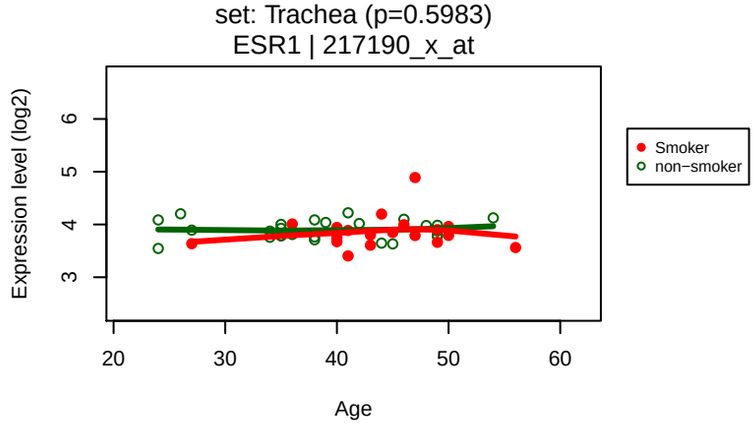

Gene expression ratio in Males
gene: ESR1 | 217190_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4503
p=0.0852
p=0.6571
p=0.9004
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ESR1 | 217190_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1001
p=0.9093
p=0.4752
p=0.3972
SmallAirway2
LargeAirway2
Trachea
Macrophage2

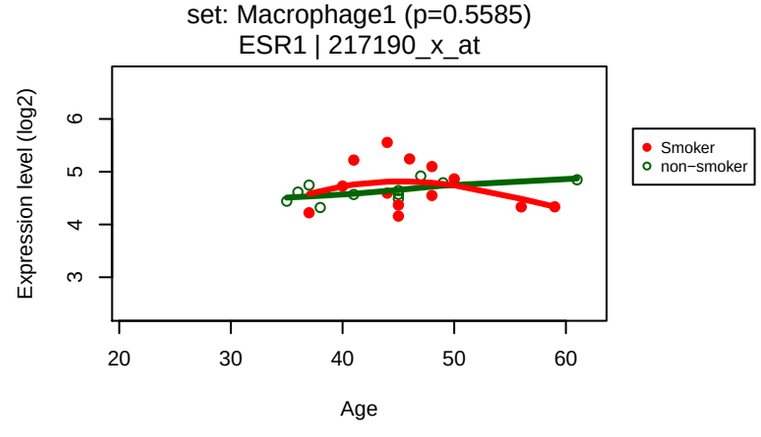

set: SmallAirway1 (p=0.147)
ESR1 | 217190_x_at
Expression level (log2)
Age
Smoker
non−smoker

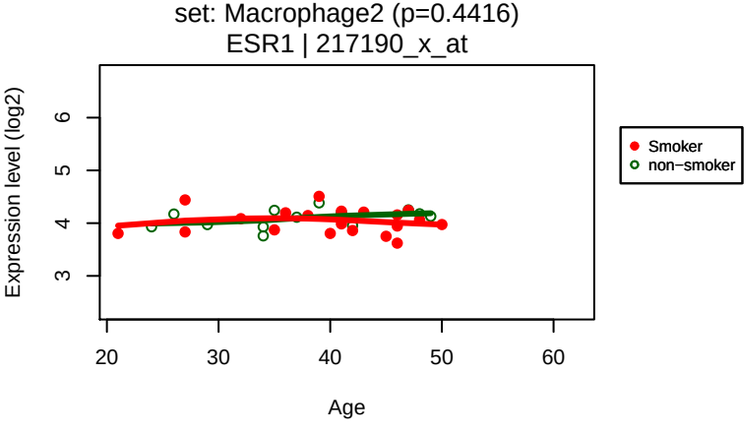

set: SmallAirway2 (p=0.096)
ESR1 | 217190_x_at
Expression level (log2)
Age
Smoker
non−smoker

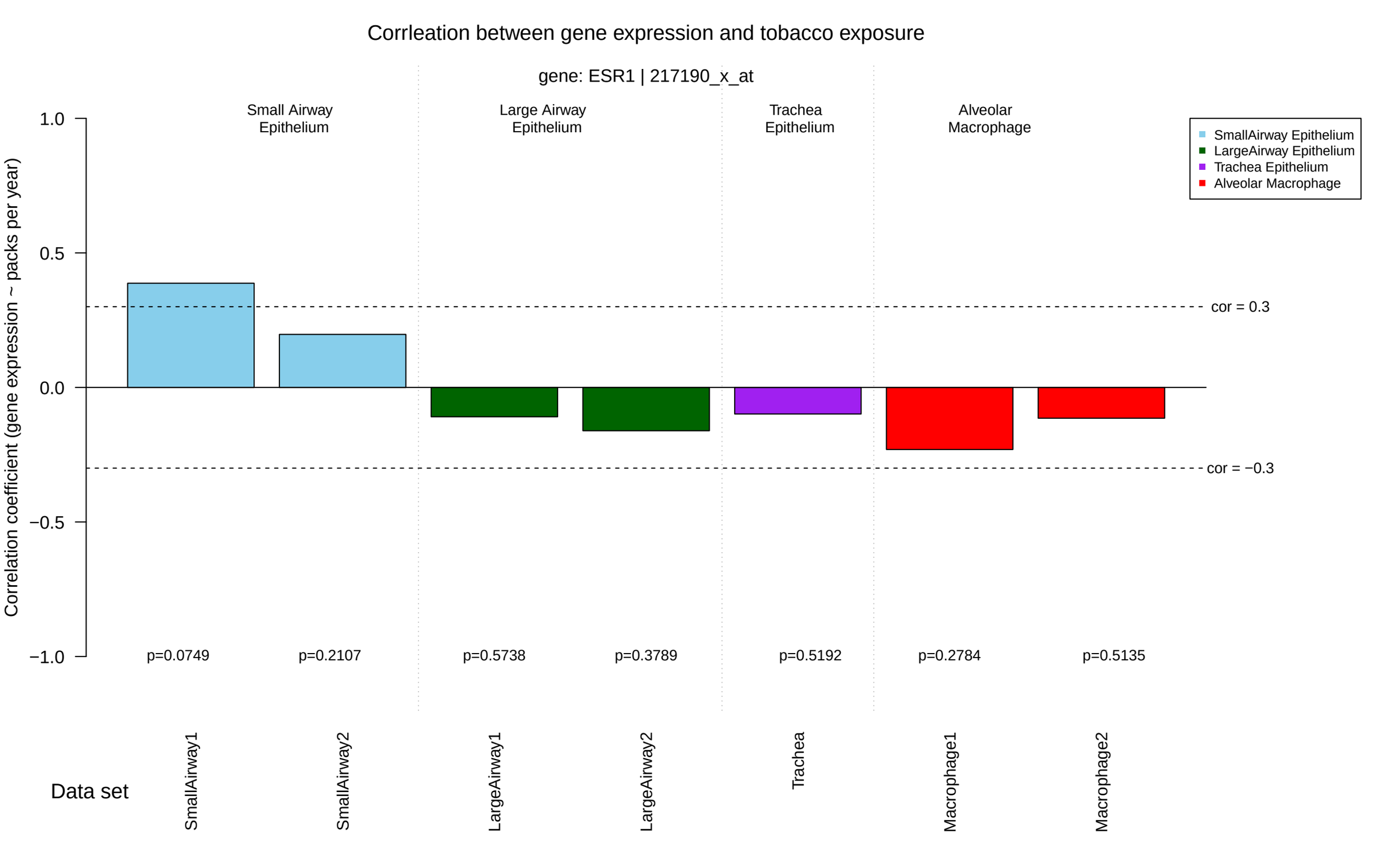

set: LargeAirway1 (p=0.1762)
ESR1 | 217190_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1183)
ESR1 | 217190_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5983)
ESR1 | 217190_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5585)
ESR1 | 217190_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4416)
ESR1 | 217190_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ESR1 | 217190_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0749
p=0.2107
p=0.5738
p=0.3789
p=0.5192
p=0.2784
p=0.5135
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2